BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Procedures and Variable Scope

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What distinguishes a function from a procedure?

2. What is SMTP used for?
   A) Downloading web pages
   B) Sending email
   C) Transferring file archives
   D) Looking up domain names

3. "Cloud storage guarantees that files remain accessible during every Internet outage." Correct this. Explain why.

4. A school buys network devices from different makers. "Standards mean all devices must have the same maker." Correct this. How do standards help?

Today's objectives

Use procedures, parameters and local and global variables. Use the technique in a different problem.

Key vocabulary - Read, cover, recall

Procedure
A named subprogram that performs a task without returning a function value.

Parameter
A named input to a procedure or function.

Local variable
A variable created inside a subprogram, ordinarily limited to its scope.

Global variable
A variable declared for use across program sections, which should be handled carefully.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Procedure
A named subprogram that performs a task without returning a function value. In the welcome message, identify where this technique applies.

Parameter
A named input to a procedure or function. In the ticket confirmation, identify where this technique applies.

Local variable
A variable created inside a subprogram, ordinarily limited to its scope. In the club announcement, identify where this technique applies.

Global variable
A variable declared for use across program sections, which should be handled carefully. In the text output routine, identify where this technique applies.

I Do - Worked model

A welcome message uses procedure and parameter. Explain what each contributes. [4 marks]

- Identify the program's purpose and the pieces of data needed.
- Procedure [1]; A named subprogram that performs a task without returning a function value. [1].
- Parameter [1]; A named input to a procedure or function. [1].
- Each feature needs a correct technical fact plus an explanation.

We Do - Guided practice

1. A ticket confirmation needs procedure. Explain the purpose. [2 marks]

2. Explain parameter for a club announcement. [2 marks]

3. When programming a text output routine, explain why local variable and global variable matter. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the call that displays a greeting for "Mia".

```
procedure greet(person)
print("Hello " + person)
endprocedure
```

You Do - Independent exam questions

Q2. State TWO relevant ideas or operations from this topic. [2 marks]

Q3. Explain parameter and local variable in a sports registration. [4 marks]

Q4. A programmer makes a library reminder. Explain the roles of procedure, local variable and global variable. [6 marks]

Exit Ticket - True or False?

A procedure is a named subprogram that performs a task without returning a value.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. A function returns a value to the caller. A procedure performs a task without returning a function value.

2. B - Sending email

3. Storage, software or processing on remote servers, accessed over a network. Access depends on the connection and service.

4. Agreed standards let hardware and software from different makers work together and exchange data.

I Do - Explain the model

Identify the program's purpose and the pieces of data needed.

Procedure [1]; A named subprogram that performs a task without returning a function value. [1].

Parameter [1]; A named input to a procedure or function. [1].

Each feature needs a correct technical fact plus an explanation.

We Do - Model answers

1. Procedure [1]; A named subprogram that performs a task without returning a function value. [1].

2. Parameter [1]; A named input to a procedure or function. [1].

3. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1].

Code First - Answer

```
procedure greet(person)
print("Hello " + person)
endprocedure
greet("Mia")
```

Expected output: Hello Mia

You Do - Self-mark

Q2: Procedure [1]; Parameter [1].

Q3: Parameter [1]; A named input to a procedure or function. [1]. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1].

Q4: Procedure [1]; A named subprogram that performs a task without returning a function value. [1]. Local variable [1]; A variable created inside a subprogram, ordinarily limited to its scope. [1]. Global variable [1]; A variable declared for use across program sections, which should be handled carefully. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.